Internet of Things (IoT)

Chapter 1: Introduction, Applications, Building Blocks, Protocols & Communication Models

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Compiled study notes for exam preparation

Contents

1. Introduction to Internet of Things

The Internet of Things (IoT) is a platform where everyday physical devices are connected to the internet and are able to exchange information with one another without needing constant human involvement. In simple words, IoT is a network of interconnected devices that can collect, share and act upon data automatically. Building such a network typically requires electronic components, internet connectivity, suitable software and supporting hardware.

The word "Things" in Internet of Things refers to any everyday object that can be connected to, or accessed through, the internet — a fan, a car, a wearable band, a refrigerator, and so on. The term "Internet of Things" is credited to Kevin Ashton, who coined it in the year 1999.

Advantages and Disadvantages of IoT

IoT Advantages	IoT Disadvantages
Better customer engagement	Security and privacy risks
Speed and automation of tasks	Increased system complexity
Easy machine-to-machine (M2M) communication	Possible unemployment due to automation
Improved and enhanced data collection	Over-dependency on technology

2. Characteristics of IoT

IoT refers to devices that are connected to the internet and are capable of sending and receiving data over a wireless network. The main characteristics of an IoT system are described below.

i) Connectivity

Connectivity is the most important characteristic of IoT. Devices and their components — such as sensors, compute engines and data hubs — remain connected to one another. Devices can connect using radio waves, Bluetooth, Wi-Fi, Li-Fi and similar mediums.

ii) Intelligence

IoT systems use a range of algorithms, along with Big Data Analytics and Machine Learning, to make better decisions. Based on the situation, the collected data is analysed to make useful business decisions — which is why IoT is considered intelligent and smart.

iii) Sensing

IoT is not possible without sensors. Sensors detect and measure changes occurring in the surrounding environment. Since sensors typically read analog signals, this data must be processed to generate meaningful insights. Depending on the requirement, different types of sensors are used — such as electrochemical sensors, gyroscopes, pressure sensors, light sensors, GPS and RFID.

iv) Dynamic Nature

IoT devices collect and convert data continuously so that useful business decisions can be made. For this, the nature of IoT components must be dynamic, i.e. their state should keep changing based on conditions. For example, a temperature sensor must adjust its reading based on location and weather — a sensor in Delhi will show a different temperature than one in Mumbai.

v) Scale

IoT systems can be scaled up or down depending on the requirement. A home IoT setup will be small in scale, while an IoT system used in a factory or large company will be much bigger. IoT infrastructure is designed keeping this scalability requirement in mind.

vi) Security

Security is a major concern in IoT because devices handle a lot of sensitive information, which can be manipulated or stolen by hackers/attackers. While designing an IoT system, security measures such as installing firewalls and using VPNs must be implemented.

vii) Communication

Devices connect with one another to communicate, and this communication can occur over both short and long distances — for example using Wi-Fi for short range or LPWA (Low Power Wide Area) networks for longer range.

viii) Scalability

The number of connected devices ("things") in an IoT infrastructure keeps increasing every day. Therefore, an IoT setup must be designed in a way that it can handle large-scale expansion.

3. How IoT Works

The complete IoT process usually begins with everyday devices such as smartphones, smart watches and electronic appliances like TVs or washing machines, which help the user communicate with the IoT platform.

Four Fundamental Components of an IoT System

- **Sensors / Devices:** A sensor or device is the front-end component that collects live data from the surrounding environment. This can range from a simple temperature sensor to a complex video feed. A single device can also contain multiple sensors performing different functions — e.g. a mobile phone has GPS, a camera and several other sensors.
- **Connectivity:** All the collected data needs to be sent to a cloud infrastructure. Various communication mediums such as satellite networks, Bluetooth, Wi-Fi and WAN are used to connect sensors to the cloud.
- **Data Processing:** Once data reaches the cloud, software processes it. This could be something simple, like checking temperature and controlling an AC or heater, or something complex, like using computer vision on video data to identify objects.
- **User Interface:** The processed information must be made available to end users — for example through an alarm on their phone, or a notification sent via email or text message. Users sometimes require an interface that actively monitors the IoT system. In more complex systems, users can also perform an action that produces a cascading effect — for instance, adjusting the temperature of a refrigerator remotely through a mobile app after being alerted of a change.

4. IoT Applications

Wearables

IoT is widely used in wearable devices such as smart watches, heart-rate monitors, glucose monitors and GPS tracking belts. Since these devices are small, they consume less energy, and they contain sensors that continuously collect data.

Healthcare

IoT is very useful for both patients and doctors in the healthcare sector. Hospitals use smart beds fitted with sensors that observe a patient's temperature, blood pressure, oxygen levels and more.

Traffic Monitoring

IoT helps monitor traffic by detecting vehicle speed and analysing traffic flow. If a vehicle does not follow traffic rules, it can be identified by a computer system and challaned (fined) automatically.

Agriculture

IoT is used to check the quality of crops and farmland. Sensors placed in the field test soil moisture, acidity, nutrients and humidity. Analysing this data helps improve crop quality and prevents future losses through early decision-making.

Smart Home

Most home appliances today have become smart, such as ACs, smart TVs, refrigerators, LED bulbs, fans, smart doors and washing machines — making daily life far more convenient.

Smart City

IoT covers many areas within a smart city, including traffic control, waste management, water distribution, electricity management and pollution monitoring.

Industrial Automation

In any industry, producing goods quickly and at lower cost is essential. IoT is extremely useful here, as automation (performing tasks without human interaction) can be achieved through it.

Surveillance

IoT is used for monitoring homes, offices, airports, railway stations, etc. CCTV cameras have special sensors that can catch unusual activity and send an alert message to the owner.

5. Building Blocks of IoT

An IoT system is built using four main blocks: Sensors, Processor, Gateway and Application. Each of these plays a specific role in creating a useful IoT system.

Sensors

Sensors act as the front end of an IoT device. Their main job is to receive necessary data from the surroundings in real time and pass it on to a database or processing system. Different sensors serve different purposes — for example, gas sensors, water quality sensors, motion sensors, moisture sensors and image sensors.

Processor

The processor is the "brain" of the IoT system, similar to a computer's CPU. Its main task is to process the raw data collected by sensors and convert it into meaningful information. It can be controlled easily by applications, and it is also responsible for securing data through encryption and decryption. Examples include microcontrollers and embedded hardware devices.

Gateway

The gateway's main function is to route the processed data and transfer it to the appropriate database or network storage. In other words, the gateway assists in the communication of data. Communication and network connectivity are essential requirements for any IoT system — examples include LAN, WAN and PAN.

Application

The application is the other end of an IoT system. It makes proper use of all the collected data and provides an interface for users to interact with that data. These can be cloud-based applications responsible for presenting the collected data, and are controlled by the user. Examples include smart home apps, security system control applications, and industrial control hub applications.

In short: raw data collected by a sensor is transferred to the processor, which converts it into meaningful information; this is then sent, via the gateway's connectivity, to a remote cloud application or database system.

6. IoT Ecosystem and its Components

An IoT ecosystem is a network of different types of devices that analyse data and communicate with one another over a network. In an IoT ecosystem, user-facing smart devices such as smartphones, tablets and sensors send commands or requests to network-connected devices. After analysing the request, the devices send back the required information to the user.

Components of the IoT Ecosystem

The IoT ecosystem includes all the elements that allow businesses, governments and consumers to connect with their IoT devices. IoT is not only changing connectivity between devices and objects, but is also enabling people to access things remotely with ease. Key components include the gateway, dashboard, network, analytics, cloud, data storage, user interface, database and ongoing development.

- **Gateway:** An important component that manages internet traffic between IoT devices and the connected network. Since different IoT devices work on different network protocols, the gateway efficiently manages this traffic — internet applications commonly rely on TCP/IP based protocols.
- **Analytics:** Analytics software analyses the data generated by IoT devices and stores the results in a cloud database. Large companies collect bulk data and analyse it to identify future opportunities for business growth.
- **Cloud:** Large-scale, industry-grade IoT solutions need to manage and process massive amounts of raw data. Cloud-based architecture is commonly used to fulfil this requirement, helping companies collect bulk data from devices and applications.
- **User Interface:** Provides a visible, physical way for the user to access the system easily. Developers aim to make interfaces user-friendly so they can be accessed without extra effort, enabling smooth interaction.
- **Dashboard:** Displays information about the IoT ecosystem to the user and enables them to control it — typically through a remote unit such as a mobile application on a smartphone, tablet, PC or smart watch.
- **Database:** A proper database system is required to store and manage the data collected from different devices and end-users.
- **Development:** IoT technology keeps evolving, and the need for further development keeps growing with time.

7. Types of Networks

Networking has developed at every level of society, from local to global. Before understanding the finer details of IoT connectivity protocols, it helps to know a few basic types of computer networks: BAN, PAN, HAN, NAN, LAN, MAN, WAN, VPN, CAN and GAN.

BAN (Body Area Network)

An interconnection of computing devices worn on, or implanted in, a person's body. A BAN usually includes a smartphone (in a pocket or bag) acting as a mobile data hub — it receives user data and transfers it to a remote database or other systems. Also known as WBAN (Wireless Body Area Network).

PAN (Personal Area Network)

One of the smallest network types, commonly connecting devices like a laptop, printer, media systems and CCTV using Bluetooth or Wi-Fi. A PAN can be as small as a single room or house and is also known as WPAN.

LAN (Local Area Network)

A network used to connect two or more computers, typically confined to a single room or building, with a range of roughly up to 1 kilometre. Also called WLAN, and defined by the IEEE 802.11 standard.

MAN (Metropolitan Area Network)

Bigger than a LAN but smaller than a WAN. It is confined to a town or city and usually connects many LANs together, e.g. cable TV networks. Coaxial cables and fibre optic cables are used to connect a MAN, and it can be public or private.

WAN (Wide Area Network)

A digital communication system used to connect cities, countries and continents — the internet being the biggest example. Data transfer speed on a WAN is roughly 10 times slower than a LAN. Microwave stations or communication satellites are used to link WAN networks.

VPN (Virtual Private Network)

A virtual communication network that logically uses the basic facilities of a computer network. It encrypts the user's connection, restricting access to the network while allowing use from remote locations.

GAN (Global Area Network)

Covers an unlimited geographical area. It is used to support an unrestricted number of mobile devices across wireless LANs, satellite coverage areas and more.

8. Technologies and Protocols in the IoT Ecosystem

IoT mainly relies on standard protocols and networking technologies. Some of the key enabling technologies of IoT are: Bluetooth, BLE, Wi-Fi, Li-Fi, Cellular Network, Z-Wave, RFID, NFC, ZigBee, LoRaWAN, 6LoWPAN, GSM, GPRS and LTE.

Bluetooth

A short-range wireless technology used to exchange data between devices within roughly 30 feet, using ultra high frequency (UHF) radio waves. It forms a personal area network. Frequency: 2.45 GHz. Maximum data transfer rate: 2.1 Mb/s. Compatible with PCs, smartphones, gaming consoles and audio devices. Developed by the Bluetooth Special Interest Group; introduced on 7 May 1998.

BLE (Bluetooth Low Energy)

Commonly known as Bluetooth 4.0. Regular Bluetooth can handle a lot of data but consumes significant battery power. Where large volumes of data don't need to be sent/received, BLE is preferred since it is cheaper and uses less battery power. Data transfer rate: about 1 Mb/s.

Wi-Fi (Wireless Fidelity)

A wireless technology used to connect computers, tablets, smartphones and other devices to one another. It is a radio signal sent to nearby devices from a wireless router, based on the IEEE 802.11 family of standards, and is mainly a LAN technology. Data transfer rate: up to 300 Mb/s.

Li-Fi (Light Fidelity)

An emerging technology that uses Visible Light Communication (VLC) instead of radio waves to transmit data. Data is modulated by a light source (transmitter) and captured by a photodiode (receiver). Its speed can be around 100 times faster than Wi-Fi.

Cellular Network

A radio network distributed over land areas known as 'cells', each served by a fixed-location transceiver. Cellular technology forms the basis of mobile phones and exists in different generations, such as 2G, 3G and 4G (LTE).

Z-Wave

A wireless communication technology mainly used in smart home networks. It uses mesh network topology and supports up to 232 nodes in a network. Its data rate is roughly up to 100 Kbps within a range of about 30 metres. It is cheaper and simpler than ZigBee for small networks.

RFID (Radio Frequency Identification)

A wireless technology used to identify or track an object, consisting of two components: a Tag and a Reader. A tag placed on an object is detected by a receiver, identifying the object. Range is roughly up to 1 metre. RFID tags are of two types — Active RFID Tag (has its own power supply, self-dependent) and Passive RFID Tag (has no power source of its own; it generates power via electromagnetic induction from the receiver's radio wave signal, so it depends on the receiver for power).

NFC (Near-Field Communication)

A wireless technology derived from RFID. It also uses an NFC Tag and NFC Reader — data stored on the tag is read by the receiver. Maximum range is about 10 cm, with a data rate of 106, 212 or 424 Kbps.

ZigBee

A wireless technology mainly based on the IEEE 802.15.4 standard. It supports low-cost, low-power devices and is used most commonly in battery-powered systems. ZigBee supports several topologies, such as star, cluster tree and mesh. Its coverage range is about 100 metres, with a data transfer rate of up to 250 Kbps.

LoRaWAN (Long Range Wide Area Network)

A newer wireless technology supporting low-cost, low-power devices with a very large range — roughly 10 to 15 kilometres — although data transmission speed is fairly slow (300 bps to 37.5 Kbps). It uses licence-free radio frequency bands such as 169 MHz, 433 MHz, 868 MHz (Europe), 915 MHz (North America), and 865-867 MHz (India).

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks)

A wireless sensor network technology used for home automation, agricultural and industrial monitoring. It uses the IPv6 protocol. It works with IEEE 802.15.4 in the 2.4 GHz band, with an outdoor range of up to ~200 m, a data rate up to 200 kbps, and can support up to ~100 nodes.

GSM (Global System for Mobile Communication)

An open, digital cellular technology used to transmit voice and data, operating on 850 MHz, 900 MHz, 1800 MHz and 1900 MHz frequency bands. GSM was developed as a digital system using the Time Division Multiple Access (TDMA) technique.

GPRS (General Packet Radio Services)

A wireless and cellular network packet-switching protocol, forming an integrated part of GSM network switching. GPRS enables emails, multimedia messages and video calls.

LTE (Long-Term Evolution)

A 4G wireless standard, considerably faster than 3G technology, with average download speeds of 15-20 Mbps and average upload speeds of 10-15 Mbps.

IoT Protocol Stack (Layered View)

IoT protocols are commonly grouped into four layers, as shown below.

Application Layer	HTTP, CoAP, WebSockets, MQTT, XMPP, DDS, AMQP
Transport Layer	TCP, UDP
Network Layer	IPv4, IPv6, 6LoWPAN
Link Layer	802.3 - Ethernet, 802.16 - WiMax, 802.11 - Wi-Fi, 802.15.4 - LR-WPAN, 2G/3G/LTE - Cellular

HTTP (Hyper Text Transfer Protocol)

A protocol that enables secure communication between a browser and a website over the internet.

CoAP (Constrained Application Protocol)

An application-layer, web-based protocol designed for constrained devices like sensors, which typically have small memory and limited processing power. It is similar to HTTP and follows the REST (Representational State Transfer) architecture.

WebSocket

A computer communications protocol that provides full-duplex communication channels over a single TCP connection. Standardized by the IETF as RFC 6455 in 2011; the API allowing web apps to use it is known as WebSockets.

MQTT (Message Queuing Telemetry Transport)

A lightweight, open messaging protocol that lets resource-constrained network clients distribute telemetry information over low-bandwidth environments.

XMPP (Extensible Messaging and Presence Protocol)

A set of open technologies for instant messaging, presence, multi-party chat, voice/video calls, collaboration, lightweight middleware, content syndication and generalized routing of XML data.

DDS (Data Distribution Service)

An IoT protocol developed for Machine-to-Machine (M2M) communication by the Object Management Group (OMG). It enables data exchange via a publish-subscribe methodology, using a brokerless architecture (unlike MQTT and CoAP).

AMQP (Advanced Message Queuing Protocol)

A more advanced protocol than MQTT — more reliable, with better security support. It enables encrypted, interoperable messaging between organizations and applications, and is used in client/server messaging and IoT device management.

TCP (Transmission Control Protocol)

A connection-oriented communications standard that lets application programs and devices exchange messages over a network, ensuring successful delivery of packets.

UDP (User Datagram Protocol)

A connectionless protocol, less common than TCP in IoT, but appealing to manufacturers because it uses fewer network resources and doesn't need to maintain a constant connection between endpoints.

9. IoT Design Methodology

IoT design methodology works somewhat like an algorithm for building an IoT ecosystem. It consists of the following 10 steps that must be followed to achieve an accurate result:

- **Purpose & Requirements** — Define the purpose and requirements of the IoT system.
- **Process Model Specification** — Define the use cases.
- **Domain Model Specification** — Define physical entities, virtual entities, devices, resources and services in the IoT system.
- **Information Model Specification** — Define the structure (relations, attributes) of all information in the system.
- **Service Specifications** — Map the process and information model to services, and define service specifications.
- **IoT Level Specification** — Define the IoT level for the system.
- **Functional View Specification** — Map the IoT level to functional groups.
- **Operational View Specification** — Define communication options, service hosting options, storage options and device options.
- **Device & Component Integration** — Integrate devices, and develop/integrate the components.
- **Application Development** — Develop the required applications.

10. Physical Design vs Logical Design of IoT

Physical Design

The physical design of an IoT system refers to the individual node devices and the protocols they use to build a functional IoT ecosystem. Each node device performs tasks such as remote sensing, actuating and monitoring, relying on physically connected devices, and may also transmit information through various wired or wireless connections.

Logical Design

A logical design is a conceptual, abstract design. At this stage, the physical implementation details are not yet considered — only the types of information needed are defined. The process of logical design arranges data into a series of logical relationships called entities and attributes.

11. IoT Functional Blocks

An IoT system consists of several functional blocks that together provide it with the ability to identify, sense, actuate, communicate and manage. Together these blocks are: Application, Management, Services, Communication, Security and Device — with the Device block forming the base layer, and Services/Communication (supported by Management and Security) forming the middle layer, leading up to the Application layer on top.

12. IoT Communication Models

An effective communication model is useful for users and also drives business growth. IoT relies on four main communication models:

- Request-Response Model
- Publish-Subscribe Model
- Push-Pull Model
- Exclusive Pair Model

Request-Response Model

In this model, a client sends a request to a server, and the server responds to it — hence it is also called the client-server model. Whenever the server receives a request, it fetches/retrieves the relevant data and sends the response back to the client. For example, when a client requests a website in a browser, the server processes that request and returns the website content to be shown to the client.

Publish-Subscribe Model

This model involves three parties: a publisher, a broker and a consumer. Publishers collect data and publish it; the broker manages this data, and consumers subscribe to receive it. Whenever the broker receives data from a publisher, it forwards that data to all consumers who have subscribed to it.

Push-Pull Model

In this communication model, data producers push data into a queue, and data collectors pull that data from the queue. In other words, "producers push data into the queue, and consumers pull data out of the queue." The queue acts as a buffer, which helps when there is a mismatch between the rate of the producer and the consumer.

Exclusive Pair Model

This is a bi-directional, fully duplex communication model that uses a persistent connection between a client and a server. Being fully duplex means both the client and the server can send messages to each other at the same time.

13. Development Tools used in IoT

Arduino

Arduino is an open-source prototyping platform based on easy-to-use hardware and software, in the form of an electronic board. It acts like the "brain" of the system and processes data coming in from sensors. An Arduino board is essentially a small CPU containing a chip known as the microcontroller (MCU). It is used as a device to read input, control connected electronics, and convert that input into output.

Tessel 2

Tessel 2 is used to build basic IoT prototypes and applications. It is a development board with Wi-Fi capability, allowing scripts to be written in Node.js. It offers ethernet connectivity, Wi-Fi connectivity, two USB ports, one micro-USB port, 32 MB flash storage and 64 MB RAM. Additional modules can also be added, such as a camera, accelerometers, RFID and GPS. It has two processors — one runs firmware applications at high speed, while the other effectively manages power and controls input/output.

Raspberry Pi

The Raspberry Pi is a low-cost, credit-card sized computer that plugs into a computer monitor or TV and uses a standard keyboard and mouse. It is a capable little device that enables people of all ages to explore computing and learn to program in languages like Scratch and Python.

14. Practice Questions (Chapter 1)

Q1. Which is the "Central Nervous System" that looks after global services in IoT?
a) Perception Layer b) Network Layer c) Application Layer d) None of these

Q2. Which of the following is a major characteristic of IoT?
a) Heterogeneity b) Connectivity c) Safety d) All of the above

Q3. Using dashboards to connect and control IoT devices enables organizations to use ___?
a) Smartphones b) Tablet and PC c) Smart Watches d) All of the above

Q4. Which of the following is an example of a short-range wireless network?
a) WWW b) Internet c) Wi-Fi d) VPN

Q5. Which is the lowest layer of IoT architecture?
a) Gateway b) Smart Devices c) Cloud d) None of these

Q6. WiFi-ah (HaLow) is specifically designed for ___?
a) 802.11 b) Long range c) Low power d) All of the above

Q7. Which of the following protocols follows a publish-subscribe architecture?
a) CoAP b) HTTPS c) MQTT d) HTTP

Q8. Which wireless technology used in IoT consumes the least power?
a) GSM/CDMA b) Wi-Fi c) Bluetooth d) Zigbee

Q9. Which of the following is NOT a main element of IoT?
a) Things b) Security c) Process d) People

Q10. ZigBee is based on which standard?
a) IEEE 802.15.4 b) IEEE 802.5 c) IEEE 802.4.13 d) Both (a) and (b)

Q11. LoRa stands for ___?
a) Long Range Radio b) Low Radio Range c) Level Range Radio d) None of these

Q12. Which of the following methods connects IoT devices with data?
a) Network b) Automata c) Cloud d) Internet

Q13. ___ is an application layer protocol for devices with limited resources?
a) TCP/IP b) MQTT c) HMTP d) CoAP

Q14. Which of the following is an entity in IoT?
a) Business b) Government c) Consumer d) All of the above

Q15. Which are the most widely used protocols in IoT?
a) MQTT b) ZigBee c) XMPP d) All of the above

Q16. Which of the following is used as a lightweight protocol in IoT applications?
a) MQTT b) WebSocket c) HTTP d) All of the above

Q17. MQTT suite is suitable for ___?
a) Small and cheap equipment b) Low memory devices c) Low power equipment d) All of the above

Q18. 6LoWPAN is ___?
a) IPv6 over low power wireless personal area network b) IPv4 over low power wireless private area network c) IPv6 over low energy wired personal area network d) IPv4 over long range wireless private area network

Q19. Which wireless standards are used in cellular networks?
a) GSM   b) CDMA   c) LTE   d) All of the above

Q20. RFID technique includes ___?
a) Antenna   b) Transponder   c) Transceiver   d) All of the above

— End of Chapter 1 —

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